

# Appendix F

## Cross checks using 2D fits

With extensive help from Chanaka DeSilva with the fits and interpretation and from Anthony Timmins with interpretation, cross checks were done using 2D fits to ensure that the data were consistent if  $v_2$  were allowed to be a free parameter. The data were fit to a Gaussian in  $\Delta\phi$  on the near-side (the ridge), a Gaussian in  $\Delta\phi$  and  $\Delta\eta$  on the near-side (the jet-like correlation), a  $\Delta\eta$  independent  $\cos(\Delta\phi)$  term for the away-side, and a  $\Delta\eta$ -independent  $v_2$  modulated background.

This was given by:

$$\frac{d^2 N}{d\Delta\phi d\Delta\eta} = A + 2B \langle v_2^{trig} \rangle \langle v_2^{assoc} \rangle \cos 2\Delta\phi + C \cos \Delta\phi \quad (\text{F.1})$$

$$+ \frac{Y_J}{2\sigma_{\Delta\phi,J}^2 \sigma_{\Delta\eta,J}^2} e^{-\frac{\Delta\eta}{2\sigma_{\Delta\eta,J}^2}} e^{-\frac{\Delta\phi}{2\sigma_{\Delta\phi,J}^2}} \quad (\text{F.2})$$

$$+ \frac{dY_R}{d\Delta\eta} \frac{1}{\sqrt{2}\sigma_{\Delta\eta,R}^2} e^{-\frac{\Delta\phi}{2\sigma_{\Delta\phi,R}^2}}. \quad (\text{F.3})$$

This is in contrast to the form assumed to describe the near-side in our analysis:

$$\frac{d^2 N}{d\Delta\phi d\Delta\eta} = B \left( 1 + 2\langle v_2^{trig} \rangle \langle v_2^{assoc} \rangle \cos 2\Delta\phi \right) \quad (\text{F.4})$$

$$+ \frac{Y_J}{2\sigma_{\Delta\phi,J}^2 \sigma_{\Delta\eta,J}^2} e^{-\frac{\Delta\phi}{2\sigma_{\Delta\phi,J}^2}} e^{-\frac{\Delta\eta}{2\sigma_{\Delta\eta,J}^2}} \quad (\text{F.5})$$

$$+ \frac{dY_R}{d\Delta\eta} \frac{1}{\sqrt{2}\sigma_{\Delta\eta,R}^2} e^{-\frac{\Delta\eta}{2\sigma_{\Delta\eta,R}^2}}. \quad (\text{F.6})$$

We do not consider the away-side in our analysis and make no assumptions about the shape of the away-side.

The fits were done in four different centrality bins in  $Au + Au$  collisions at  $\sqrt{s_{NN}} = 200$  GeV. The jet-like yield is consistent with the method used in this paper. The widths in  $\Delta\phi$  and  $\Delta\eta$  are somewhat smaller in the fit method.  $v_2$  is smaller with the fit method. For the 0-12% bin,  $v_2$  from the fit is negative.

Qualitatively the results are the same - the ridge yield is larger in central collisions than peripheral collisions. This is the behavior we are interested in demonstrating in our paper. Quantitatively the ridge yields are quite different. For comparison with the other method, the  $\cos(\Delta\phi)$  term is added to the ridge in order to get comparable terms. The  $\cos(\Delta\phi)$  term appears to lead to a higher ridge yield. In the fit method, the  $\cos(\Delta\phi)$  term is considered to be an off-set from the  $v_2$ -modulated background, leading to a smaller background from the  $\cos(2\Delta\phi)$  term. This leads to a higher ridge yield. In addition, the ridge was fit to have a width of roughly 0.5 radians using the fit; this is likely because the dip created by the  $\cos(\Delta\phi)$  term must be compensated for by the Gaussian for the ridge.

The projections of the different terms in the fits are shown in Figure F.2. The  $\cos(\Delta\phi)$  term leads to a structure which dramatically changes shape underneath the near-side peak. It isn't clear how to interpret these shapes.

| 0-12%                                         |                                            |                    |
|-----------------------------------------------|--------------------------------------------|--------------------|
|                                               | ZYAM                                       | fit                |
| $v_2$ (trig)* $v_2$ (assoc)                   | 0.0045 (low), 0.0070 (avg.), 0.0100 (high) | -0.0021            |
| $\cos(\Delta\phi)$ term coefficient           | N/A                                        | -0.097 $\pm$ 0.001 |
| near-side $\Delta\phi$ width                  | 0.330 $\pm$ 0.006                          | 0.318 $\pm$ 0.002  |
| near-side $\Delta\eta$ width                  | 0.395 $\pm$ 0.008                          | 0.363 $\pm$ 0.003  |
| jet-like yield                                | 0.1298 $\pm$ 0.0015                        | 0.134              |
| ridge yield (width)                           | 0.270 $\pm$ 0.006 $^{+0.126}_{-0.160}$     | 0.664 (0.52731)    |
| $v_3$ *multiplicity ( $v_3/B$ ) of ridge term |                                            | 0.106 (0.0324)     |
| 10-20%                                        |                                            |                    |
|                                               | ZYAM                                       | fit                |
| $v_2$ (trig)* $v_2$ (assoc)                   | 0.0148 (low), 0.0189 (avg.), 0.0236 (high) | 0.0078             |
| $\cos(\Delta\phi)$ term coefficient           | N/A                                        | -0.109 $\pm$ 0.003 |
| near-side $\Delta\phi$ width                  | 0.304 $\pm$ 0.013                          | 0.271 $\pm$ 0.004  |
| near-side $\Delta\eta$ width                  | 0.328 $\pm$ 0.014                          | 0.310 $\pm$ 0.005  |
| jet-like yield                                | 0.1168 $\pm$ 0.0029                        | 0.114              |
| ridge yield (width)                           | 0.251 $\pm$ 0.012 $^{+0.167}_{-0.189}$     | 0.719 (0.56238)    |
| $v_3$ *multiplicity ( $v_3/B$ ) of ridge term |                                            | 0.101 (0.0438)     |
| 20-30%                                        |                                            |                    |
|                                               | ZYAM                                       | fit                |
| $v_2$ (trig)* $v_2$ (assoc)                   | 0.0242 (low), 0.0294 (avg.), 0.0352 (high) | 0.019              |
| $\cos(\Delta\phi)$ term coefficient           | N/A                                        | -0.088 $\pm$ 0.003 |
| near-side $\Delta\phi$ width                  | 0.298 $\pm$ 0.011                          | 0.273 $\pm$ 0.003  |
| near-side $\Delta\eta$ width                  | 0.321 $\pm$ 0.012                          | 0.296 $\pm$ 0.004  |
| jet-like yield                                | 0.1226 $\pm$ 0.0027                        | 0.117              |
| ridge yield (width)                           | 0.191 $\pm$ 0.011 $^{+0.149}_{-0.166}$     | 0.507 (0.56978)    |
| $v_3$ *multiplicity ( $v_3/B$ ) of ridge term |                                            | 0.0731 (0.0466)    |
| 30-40%                                        |                                            |                    |
|                                               | ZYAM                                       | fit                |
| $v_2$ (trig)* $v_2$ (assoc)                   | 0.0278 (low), 0.0343 (avg.), 0.0414 (high) | 0.029              |
| $\cos(\Delta\phi)$ term coefficient           | N/A                                        | -0.056 $\pm$ 0.003 |
| near-side $\Delta\phi$ width                  | 0.287 $\pm$ 0.012                          | 0.260 $\pm$ 0.003  |
| near-side $\Delta\eta$ width                  | 0.293 $\pm$ 0.012                          | 0.261 $\pm$ 0.003  |
| jet-like yield                                | 0.1128 $\pm$ 0.0026                        | 0.105              |
| ridge yield (width)                           | 0.187 $\pm$ 0.011 $^{+0.119}_{-0.135}$     | 0.248 (0.54699)    |
| $v_3$ *multiplicity ( $v_3/B$ ) of ridge term |                                            | 0.0456 (0.0441)    |

Table F.1: Fit parameters for 0-12% central  $Au + Au$  at  $\sqrt{s_{NN}} = 200$  GeV using both ZYAM and a fit

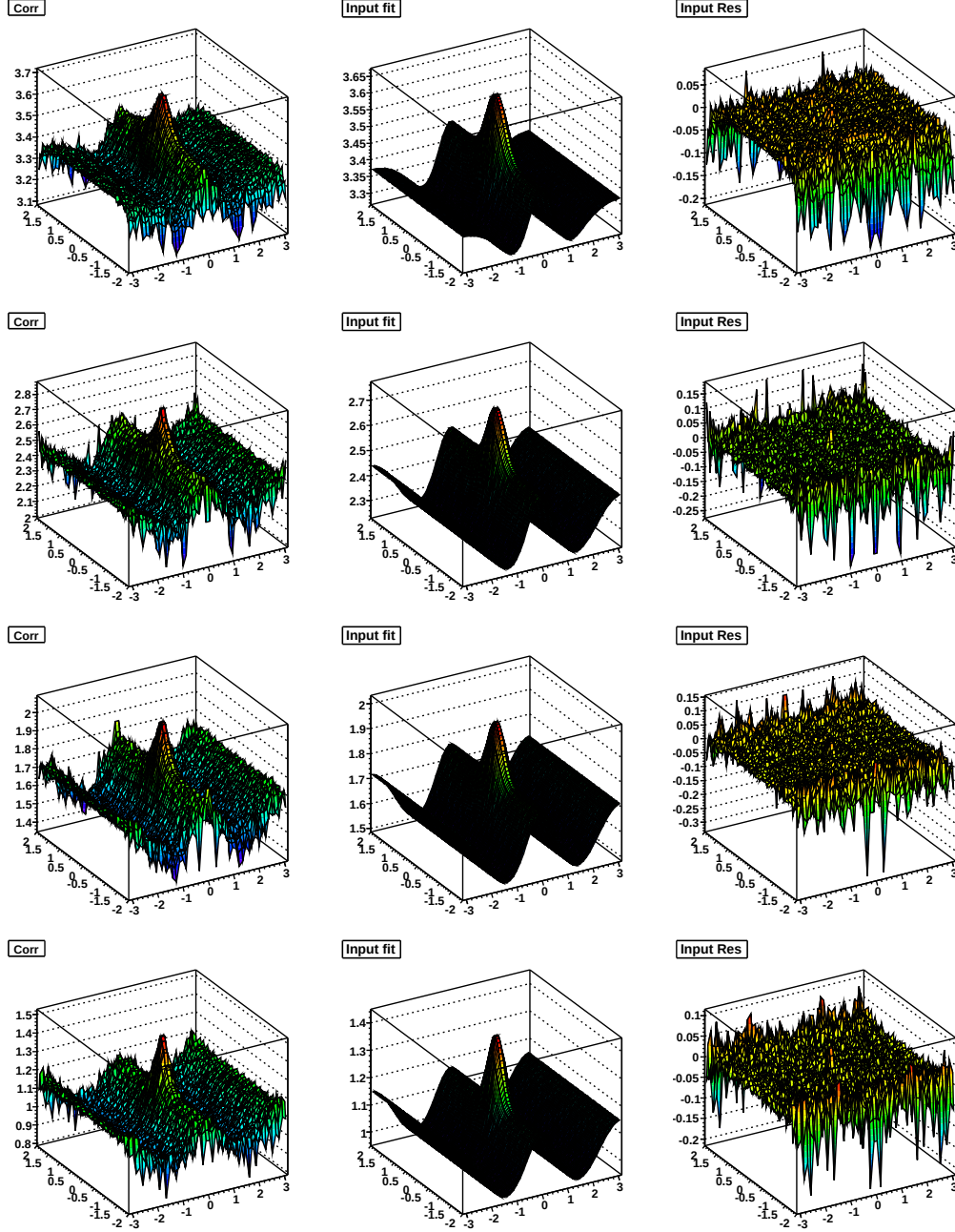


Figure F.1: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 0-12% (top), 10-20% (second down), 20-30% (third down), and 30-40% (bottom)  $Au + Au \sqrt{s_{NN}} = 200 \text{ GeV}$



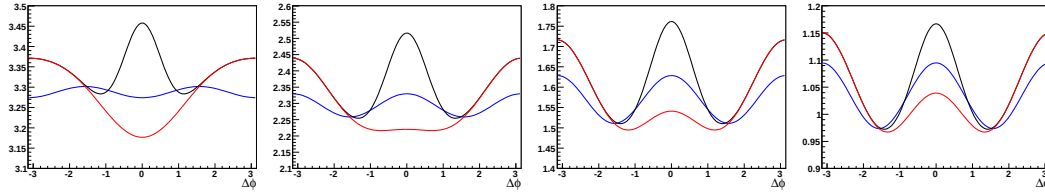


Figure F.2: Projections of ridge,  $\cos(\Delta\phi)$ , and  $v_2$  modulated background terms from fits from left to right for for 0-12%, 10-20%, 20-30%, and 30-40%  $Au + Au$   $\sqrt{s_{NN}} = 200$  GeV. The black line shows the total, the blue line shows the  $v_2$  modulated background, and the red line shows the  $v_2$  modulated background plus the  $\cos(\Delta\phi)$  term. Note that the jet-like correlation has been omitted from these projections.

### F.0.1 Extraction of $(\frac{v_3}{v_2})^2$

These fits are also used to extract  $(\frac{v_3}{v_2})^2$ . The data are fit to

$$\begin{aligned} \frac{d^2N}{d\Delta\phi d\Delta\eta} = & A \left( 1 + 2v_1^2 \cos \Delta\phi + 2v_2^2 \cos 2\Delta\phi + 2\frac{v_3^2}{v_2^2} v_2^2 \cos 3\Delta\phi + 2\frac{v_4^2}{v_2^2} v_2^2 \cos 4\Delta\phi \right) \\ & + A_J e^{-\frac{\Delta\eta}{2\sigma_{\Delta\eta,J}^2}} e^{-\frac{\Delta\phi}{2\sigma_{\Delta\phi,J}^2}}. \end{aligned} \quad (\text{F.7})$$

The output of these fits is given in Table F.2 and Table F.3 and the fits are shown in Figure F.6, Figure F.7, Figure F.8, Figure F.9, Figure F.10, and Figure F.12 for  $3.0 < p_T^{trigger} < 6.0$  GeV/c and  $1.5$  GeV/c  $< p_T^{associated} < p_T^{trigger}$ . The results from these fits were cross checked by using fits with the ridge described as a Gaussian and the  $v_3$  calculated from the Fourier decomposition of the fit function, omitting the jet-like correlation term. Proper error propagation was not done for this form as this was done mainly as a cross check. The output of these fits is given in Table F.4 and Table F.5 and the fits are shown in Figure F.13, Figure F.14, Figure F.15, Figure F.16, Figure F.17, and Figure F.18 for  $3.0 < p_T^{trigger} < 6.0$  GeV/c and  $1.5$  GeV/c  $< p_T^{associated} < p_T^{trigger}$ . Figure F.3 and Figure F.4 show the  $(\tilde{v}_1/\tilde{v}_2)^2$  and  $(\tilde{v}_4/\tilde{v}_2)^2$ , respectively, from the fit.

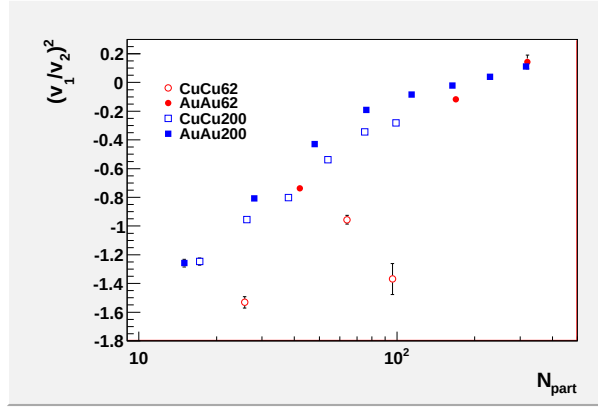


Figure F.3: The ratio of  $(\tilde{v}_1/\tilde{v}_2)^2$  from the fit.

Figure F.5 shows the  $(\frac{v_3}{v_2})^2$  determined from both functional forms. These two fit forms are numerically consistent for every data point. Comparisons are only done for the case when  $v_3^2 > 0$ . Figure F.19 shows the difference between the  $\tilde{v}_2$  determined

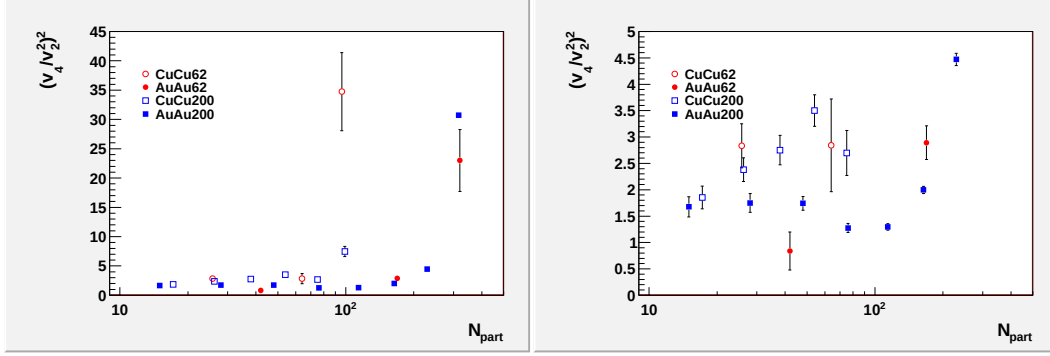


Figure F.4: The ratio of  $(\tilde{v}_4/\tilde{v}_2^2)^2$  from the fit showing both the full range (left) and zoomed (right) in to smaller values.

from the fit and the standard  $v_2^{trigger} v_2^{assoc}$  used in the analysis as a function of  $N_{part}$ . Statistical errors on the fit have been added in quadrature to the systematic errors on  $v_2^{trigger} v_2^{assoc}$ . The values used in this plot are given in Table F.6. As expected, these values are in agreement for large  $N_{part}$  and the values diverge for smaller  $N_{part}$ .

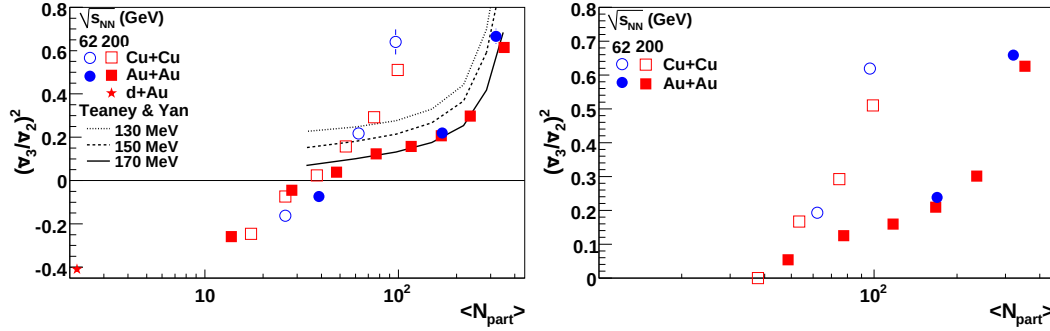


Figure F.5: Fits with the ridge described as a Gaussian (left) and as a  $v_3$  term (right).

|                                          | A      | $v_1^2$ | $v_2^2$ | $\frac{v_3^2}{v_2^2}$ | $\frac{v_4^2}{v_2^2}$ | $Y_J$  | $\sigma_{\Delta\eta,J}$ | $\sigma_{\Delta\phi,J}$ | $\chi^2/\text{NDF}$ |
|------------------------------------------|--------|---------|---------|-----------------------|-----------------------|--------|-------------------------|-------------------------|---------------------|
| <i>Cu + Cu</i> $\sqrt{s_{NN}} = 62$ GeV  |        |         |         |                       |                       |        |                         |                         |                     |
| 0-10%                                    | 0.4538 | -0.0115 | 0.0084  | 0.6396                | 0.2918                | 0.0576 | 0.3248                  | 0.3444                  | 0.87                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0003 | 0.0006  | 0.0005  | 0.0579                | 0.0533                | 0.0037 | 0.0202                  | 0.0197                  |                     |
| 10-30%                                   | 0.2698 | -0.0218 | 0.0228  | 0.2165                | 0.0648                | 0.0655 | 0.2402                  | 0.3071                  | 1.09                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0005  | 0.0005  | 0.0202                | 0.0200                | 0.0031 | 0.0106                  | 0.0122                  |                     |
| 30-60%                                   | 0.1121 | -0.0750 | 0.0471  | -0.1629               | 0.1279                | 0.0774 | 0.2268                  | 0.3092                  | 1.15                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0011  | 0.0010  | 0.0210                | 0.0201                | 0.0028 | 0.0070                  | 0.0090                  |                     |
| <i>Au + Au</i> $\sqrt{s_{NN}} = 62$ GeV  |        |         |         |                       |                       |        |                         |                         |                     |
| 0-10%                                    | 1.6515 | 0.0009  | 0.0063  | 0.6662                | 0.1449                | 0.0537 | 0.4015                  | 0.2730                  | 1.16                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0007 | 0.0003  | 0.0003  | 0.0354                | 0.0325                | 0.0083 | 0.0681                  | 0.0454                  |                     |
| 10-40%                                   | 0.8811 | -0.0030 | 0.0258  | 0.2203                | 0.0746                | 0.0596 | 0.3543                  | 0.3066                  | 0.89                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0003 | 0.0003  | 0.0002  | 0.0083                | 0.0082                | 0.0036 | 0.0224                  | 0.0164                  |                     |
| 40-80%                                   | 0.2259 | -0.0349 | 0.0456  | -0.0732               | 0.0296                | 0.0671 | 0.2721                  | 0.3252                  | 0.99                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0003 | 0.0009  | 0.0008  | 0.0177                | 0.0171                | 0.0039 | 0.0141                  | 0.0156                  |                     |
| <i>Cu + Cu</i> $\sqrt{s_{NN}} = 200$ GeV |        |         |         |                       |                       |        |                         |                         |                     |
| 0-10%                                    | 0.8962 | -0.0036 | 0.0128  | 0.5108                | 0.0958                | 0.2540 | 0.2477                  | 0.2574                  | 1.28                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0002  | 0.0002  | 0.0120                | 0.0111                | 0.0034 | 0.0031                  | 0.0030                  |                     |
| 10-20%                                   | 0.6148 | -0.0073 | 0.0212  | 0.2917                | 0.0572                | 0.2639 | 0.2386                  | 0.2623                  | 1.51                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0002  | 0.0002  | 0.0092                | 0.0090                | 0.0033 | 0.0027                  | 0.0029                  |                     |
| 10-30%                                   | 0.4207 | -0.0159 | 0.0296  | 0.1581                | 0.1036                | 0.2708 | 0.2346                  | 0.2569                  | 1.27                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0003  | 0.0003  | 0.0090                | 0.0088                | 0.0034 | 0.0024                  | 0.0028                  |                     |
| 30-40%                                   | 0.2832 | -0.0304 | 0.0379  | 0.0234                | 0.1043                | 0.2813 | 0.2238                  | 0.2567                  | 1.30                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0005  | 0.0004  | 0.0108                | 0.0105                | 0.0039 | 0.0025                  | 0.0028                  |                     |
| 40-50%                                   | 0.1861 | -0.0497 | 0.0506  | -0.0738               | 0.1182                | 0.2925 | 0.2157                  | 0.2426                  | 1.19                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0007  | 0.0006  | 0.0120                | 0.0115                | 0.0042 | 0.0023                  | 0.0027                  |                     |
| 50-60%                                   | 0.1190 | -0.0872 | 0.0657  | -0.2459               | 0.1182                | 0.2763 | 0.2174                  | 0.2670                  | 1.30                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0011  | 0.0010  | 0.0156                | 0.0146                | 0.0047 | 0.0026                  | 0.0033                  |                     |

Table F.2: Fit parameters for data in Figure F.6, Figure F.7, Figure F.8, Figure F.9, Figure F.10, and Figure F.12.

|                                          | A      | $v_1^2$ | $v_2^2$ | $\frac{v_3^2}{v_2^2}$ | $\frac{v_4^2}{v_2^2}$ | $Y_J$  | $\sigma_{\Delta\eta,J}$ | $\sigma_{\Delta\phi,J}$ | $\chi^2/\text{NDF}$ |
|------------------------------------------|--------|---------|---------|-----------------------|-----------------------|--------|-------------------------|-------------------------|---------------------|
| <i>Au + Au</i> $\sqrt{s_{NN}} = 200$ GeV |        |         |         |                       |                       |        |                         |                         |                     |
| 0-12%                                    | 3.3478 | 0.0009  | 0.0081  | 0.6146                | 0.2487                | 0.1917 | 0.3609                  | 0.2986                  | 1.59                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0001 | 0.0000  | 0.0000  | 0.0028                | 0.0026                | 0.0011 | 0.0028                  | 0.0017                  |                     |
| 10-20%                                   | 2.3607 | 0.0009  | 0.0225  | 0.2976                | 0.1006                | 0.2232 | 0.3116                  | 0.2589                  | 1.08                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0001  | 0.0001  | 0.0027                | 0.0026                | 0.0030 | 0.0046                  | 0.0033                  |                     |
| 20-30%                                   | 1.6200 | -0.0007 | 0.0344  | 0.2070                | 0.0687                | 0.2342 | 0.2973                  | 0.2667                  | 1.16                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0001  | 0.0001  | 0.0024                | 0.0023                | 0.0029 | 0.0038                  | 0.0029                  |                     |
| 30-40%                                   | 1.0621 | -0.0036 | 0.0430  | 0.1571                | 0.0557                | 0.2509 | 0.2618                  | 0.2547                  | 1.13                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0001  | 0.0001  | 0.0027                | 0.0027                | 0.0032 | 0.0031                  | 0.0029                  |                     |
| 40-50%                                   | 0.6639 | -0.0088 | 0.0461  | 0.1235                | 0.0588                | 0.2694 | 0.2544                  | 0.2518                  | 1.08                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0002  | 0.0002  | 0.0040                | 0.0039                | 0.0034 | 0.0029                  | 0.0029                  |                     |
| 50-60%                                   | 0.3901 | -0.0205 | 0.0478  | 0.0387                | 0.0833                | 0.2846 | 0.2289                  | 0.2525                  | 1.05                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0003  | 0.0003  | 0.0063                | 0.0062                | 0.0037 | 0.0024                  | 0.0029                  |                     |
| 60-70%                                   | 0.2126 | -0.0450 | 0.0546  | -0.0451               | 0.0933                | 0.2908 | 0.2250                  | 0.2500                  | 0.97                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0006  | 0.0006  | 0.0102                | 0.0098                | 0.0042 | 0.0025                  | 0.0030                  |                     |
| 70-80%                                   | 0.1066 | -0.1018 | 0.0754  | -0.2587               | 0.1255                | 0.2948 | 0.2263                  | 0.2494                  | 1.03                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0013  | 0.0012  | 0.0162                | 0.0148                | 0.0053 | 0.0029                  | 0.0031                  |                     |
| <i>d + Au</i> $\sqrt{s_{NN}} = 200$ GeV  |        |         |         |                       |                       |        |                         |                         |                     |
| MB                                       | 0.0923 | -0.1300 | 0.0694  | -0.4105               | 0.1332                | 0.2689 | 0.2469                  | 0.2583                  | 1.20                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0016  | 0.0014  | 0.0223                | 0.0194                | 0.0036 | 0.0030                  | 0.0029                  |                     |

Table F.3: Fit parameters for data in Figure F.6, Figure F.7, Figure F.8, Figure F.9, Figure F.10, and Figure F.12.

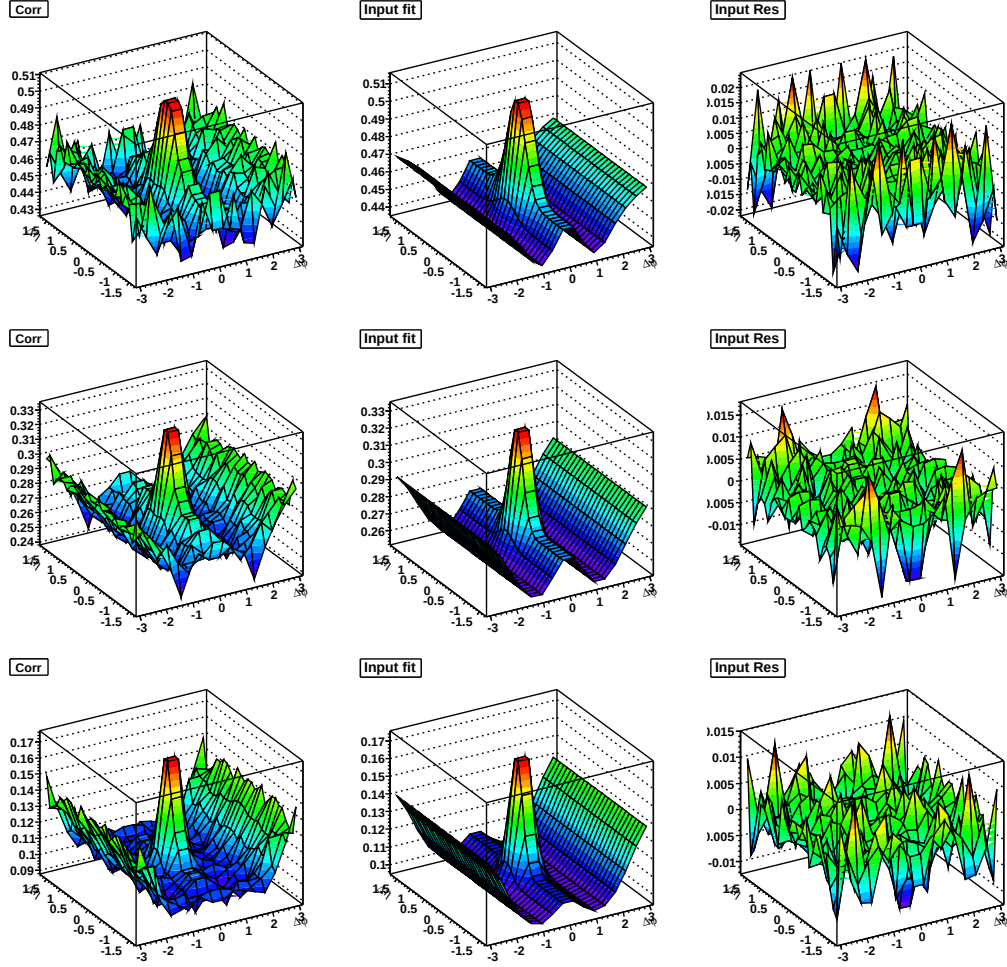


Figure F.6: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 0-10% (top), 10-30% (middle), and 30-60% (bottom)  $Cu + Cu \sqrt{s_{NN}} = 62 \text{ GeV}$

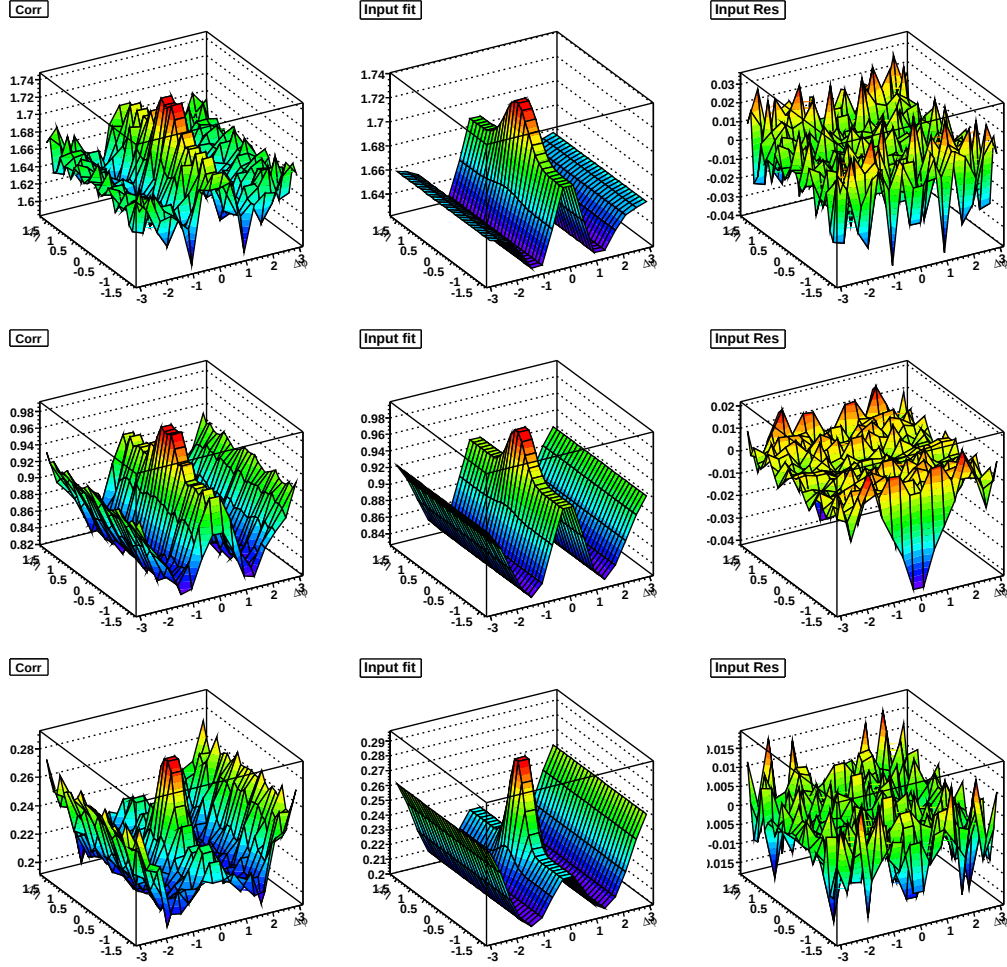


Figure F.7: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 0-10% (top), 10-40% (middle), and 40-80% (bottom)  $Au + Au \sqrt{s_{NN}} = 62 \text{ GeV}$

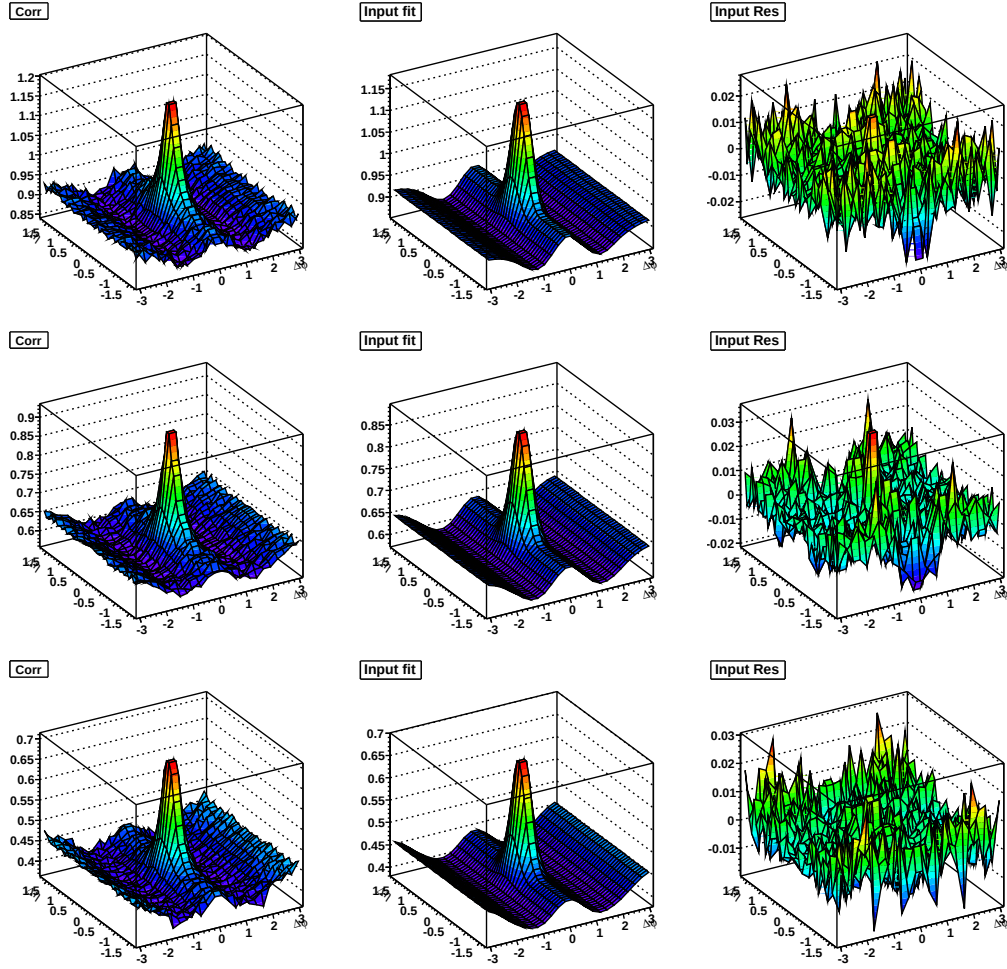


Figure F.8: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 0-10% (top), 10-20% (middle), and 20-30% (bottom)  $\text{Cu} + \text{Cu}$   $\sqrt{s_{NN}} = 200 \text{ GeV}$



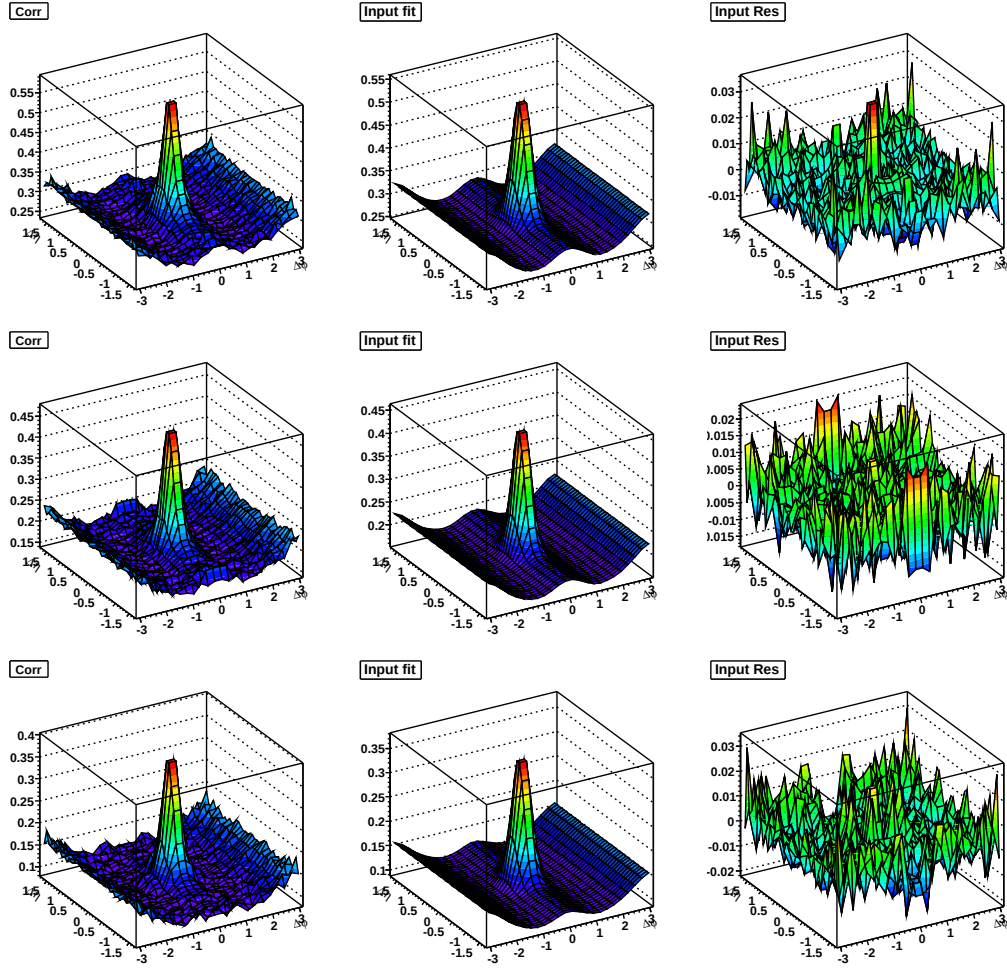


Figure F.9: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV}/c < p_T^{associated} < p_T^{trigger}$  and  $3.0 < p_T^{trigger} < 6.0 \text{ GeV}/c$  in 30-40% (top), 40-50% (middle), and 50-60% (bottom)  $Cu + Cu$   $\sqrt{s_{NN}} = 200 \text{ GeV}$

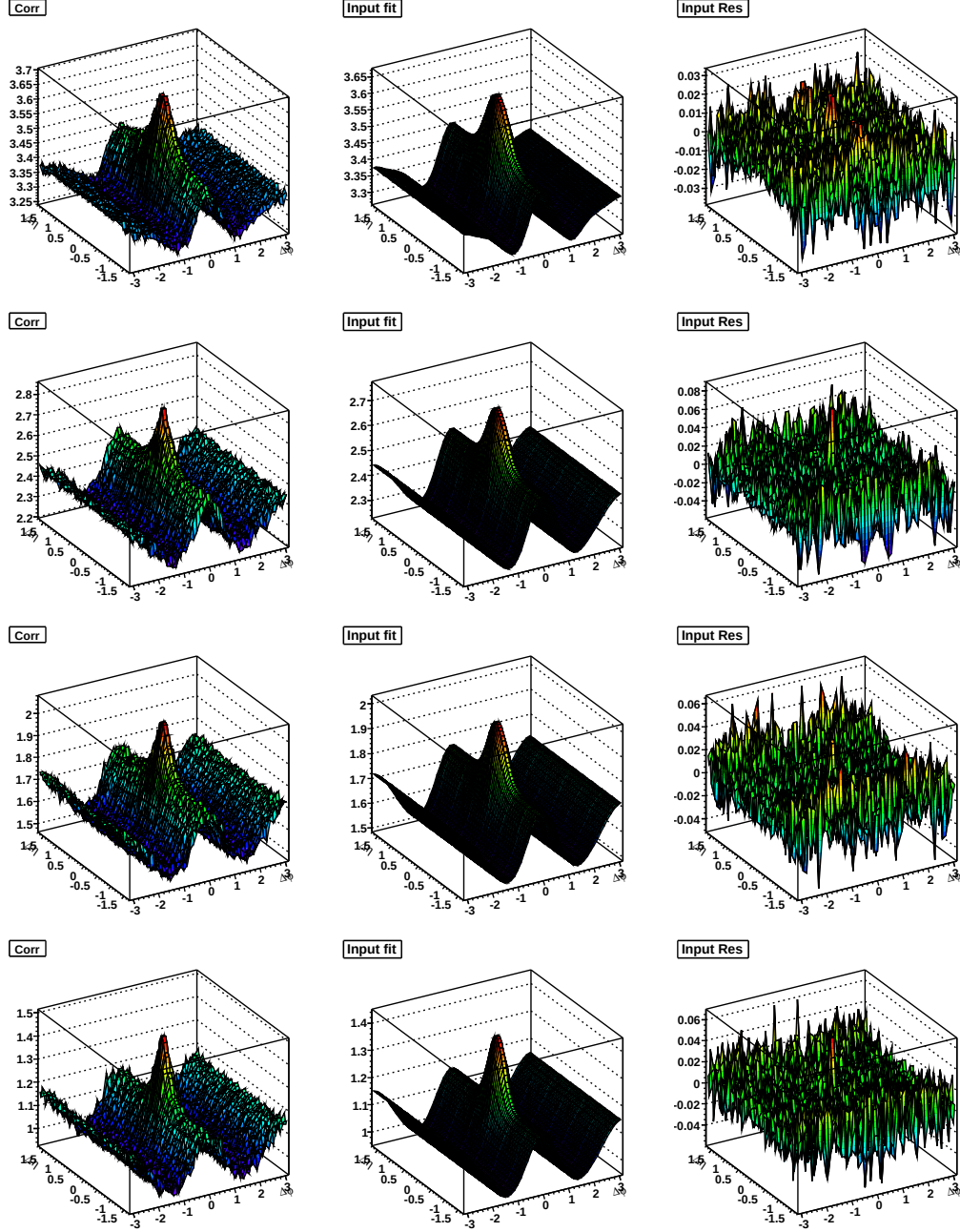


Figure F.10: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 0-12% (top), 10-20% (second down), 20-30% (third down), and 30-40% (bottom)  $Au + Au \sqrt{s_{NN}} = 200 \text{ GeV}$

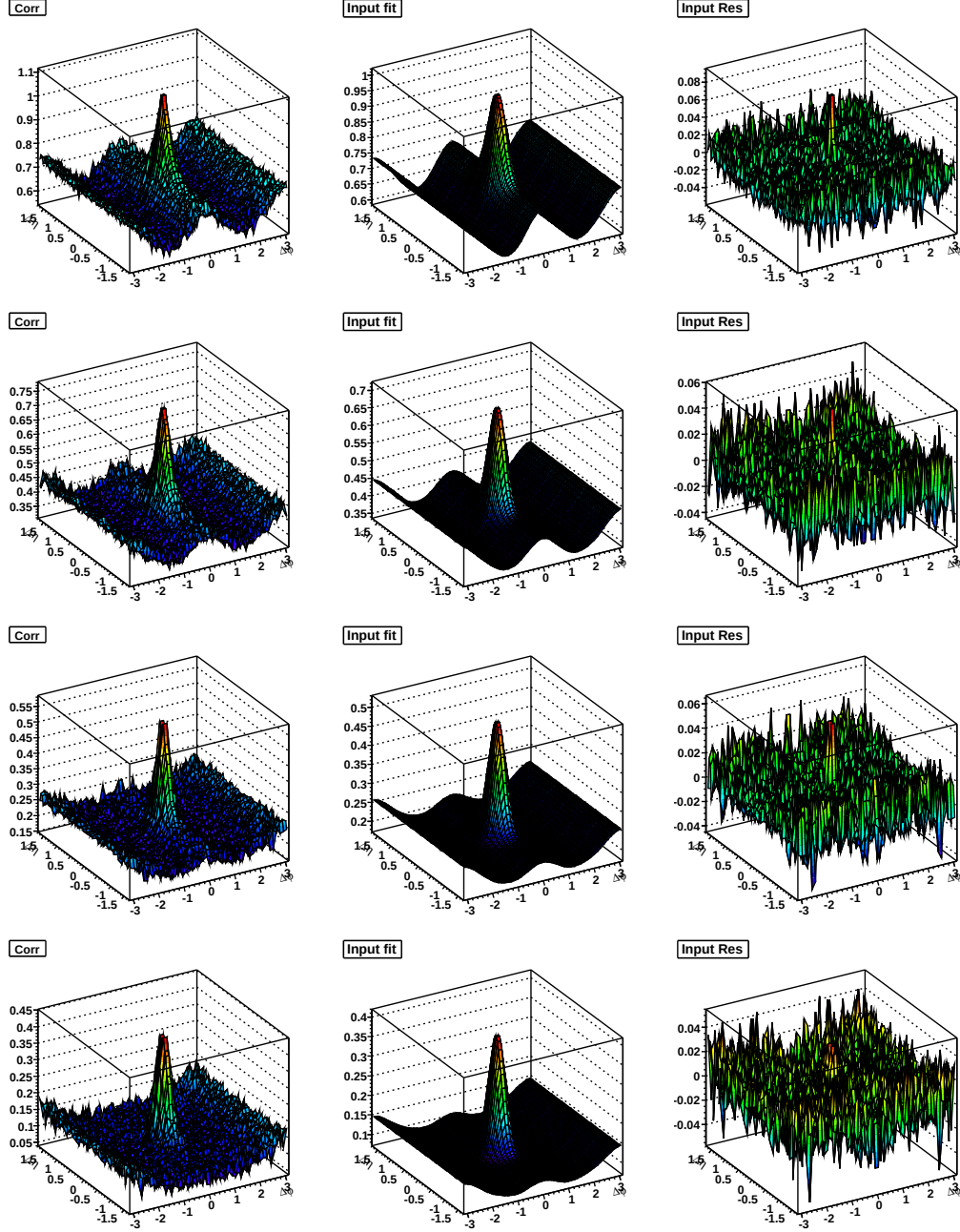


Figure F.11: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 40-50% (top), 50-60% (second down), 60-70% (third down), and 70-80% (bottom)  $Au + Au \sqrt{s_{NN}} = 200 \text{ GeV}$

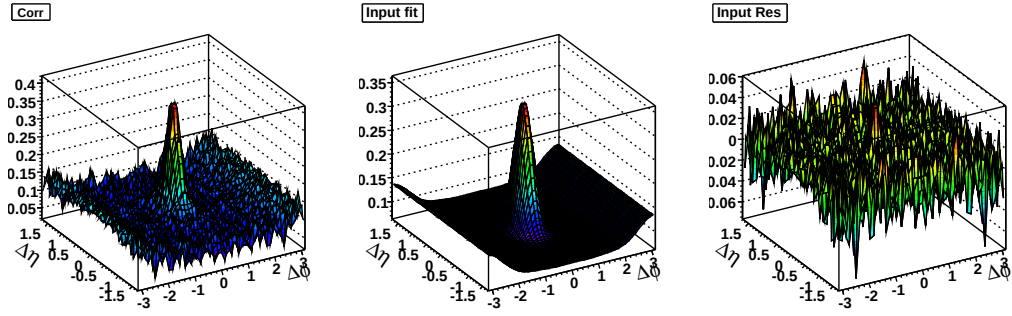


Figure F.12: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV}/c < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV}/c$  in minimum bias  $d + Au$   $\sqrt{s_{NN}} = 200 \text{ GeV}$

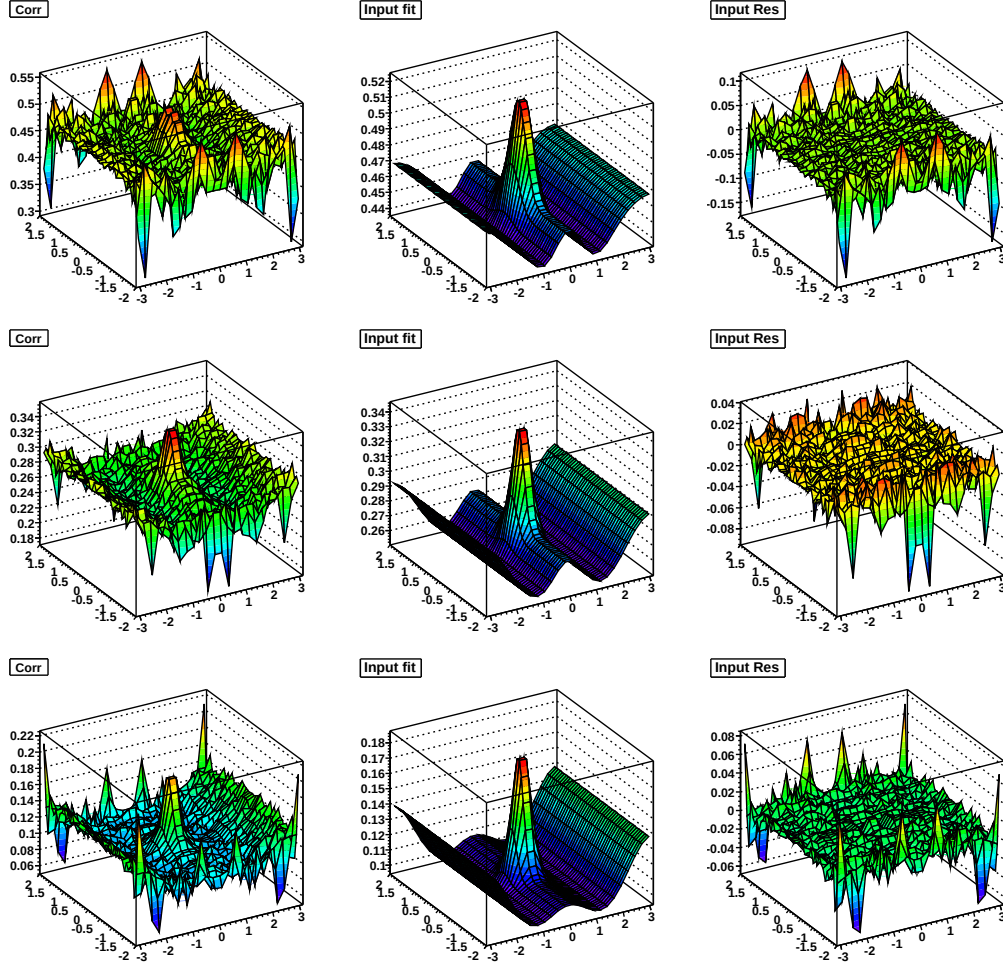


Figure F.13: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 0-10% (top), 10-30% (middle), and 30-60% (bottom)  $\text{Cu} + \text{Cu}$   $\sqrt{s_{NN}} = 62 \text{ GeV}$

|                                           | A      | $v_1^2$ | $v_2^2$ | $\frac{dY_B}{d\Delta\eta}$ | $\sigma_{\Delta\phi,R}$ | $Y_J$  | $\sigma_{\Delta\eta,J}$ | $\sigma_{\Delta\phi,J}$ | $\chi^2/\text{NDF}$ |
|-------------------------------------------|--------|---------|---------|----------------------------|-------------------------|--------|-------------------------|-------------------------|---------------------|
| $Cu + Cu \sqrt{s_{NN}} = 62 \text{ GeV}$  |        |         |         |                            |                         |        |                         |                         |                     |
| 0-10%                                     | 0.4422 | -0.0296 | -0.0036 | 0.0487                     | 0.5839                  | 0.0670 | 0.3204                  | 0.2982                  | 1.04                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0036 | 0.0054  | 0.0022  | 0.0109                     | 0.0547                  | 0.0038 | 0.0185                  | 0.0149                  |                     |
| 10-30%                                    | 0.2639 | -0.0214 | 0.0066  | 0.0247                     | 0.5936                  | 0.0738 | 0.2319                  | 0.2774                  | 1.07                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0020 | 0.0031  | 0.0012  | 0.0062                     | 0.0566                  | 0.0033 | 0.0094                  | 0.0105                  |                     |
| 30-60%                                    | 0.1121 | -0.0163 | 0.0107  | 0.0012                     | 0.0000                  | 0.0829 | 0.2171                  | 0.2795                  | 1.10                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0002 | 0.0002  | 0.0002  | 1.4223                     | 1.0649                  | 0.0032 | 0.0065                  | 0.0099                  |                     |
| $Au + Au \sqrt{s_{NN}} = 62 \text{ GeV}$  |        |         |         |                            |                         |        |                         |                         |                     |
| 0-10%                                     | 1.6210 | -0.0485 | -0.0104 | 0.1346                     | 0.5686                  | 0.0564 | 0.3286                  | 0.2867                  | 1.11                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0052 | 0.0079  | 0.0033  | 0.0159                     | 0.0304                  | 0.0072 | 0.0469                  | 0.0280                  |                     |
| 10-40%                                    | 0.8545 | -0.0497 | 0.0198  | 0.1137                     | 0.5934                  | 0.0684 | 0.2967                  | 0.2761                  | 0.97                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0036 | 0.0054  | 0.0021  | 0.0107                     | 0.0232                  | 0.0044 | 0.0181                  | 0.0159                  |                     |
| 40-80%                                    | 0.2260 | -0.0153 | 0.0206  | 0.0000                     | 1.2698                  | 0.0665 | 0.2620                  | 0.3240                  | 1.09                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0003 | 0.0020  | 0.0004  | 0.0251                     | 0.7701                  | 0.0039 | 0.0127                  | 0.0158                  |                     |
| $Cu + Cu \sqrt{s_{NN}} = 200 \text{ GeV}$ |        |         |         |                            |                         |        |                         |                         |                     |
| 0-10%                                     | 0.8379 | -0.0968 | -0.0190 | 0.2035                     | 0.7164                  | 0.2536 | 0.2495                  | 0.2585                  | 1.25                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0091 | 0.0125  | 0.0037  | 0.0250                     | 0.0236                  | 0.0034 | 0.0032                  | 0.0030                  |                     |
| 10-20%                                    | 0.5768 | -0.0678 | -0.0013 | 0.1327                     | 0.7158                  | 0.2637 | 0.2398                  | 0.2630                  | 1.45                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0083 | 0.0114  | 0.0034  | 0.0228                     | 0.0326                  | 0.0033 | 0.0027                  | 0.0029                  |                     |
| 10-30%                                    | 0.4138 | -0.0254 | 0.0167  | 0.0333                     | 0.5160                  | 0.2708 | 0.2354                  | 0.2573                  | 1.25                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0010 | 0.0016  | 0.0008  | 0.0032                     | 0.0264                  | 0.0034 | 0.0024                  | 0.0028                  |                     |
| 30-40%                                    | 0.2829 | -0.0177 | 0.0209  | 0.0025                     | 0.0003                  | 0.2837 | 0.2297                  | 0.2546                  | 1.36                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0002 | 0.0003  | 0.0002  | 1.4572                     | 1.0766                  | 0.0038 | 0.0024                  | 0.0027                  |                     |
| 40-50%                                    | 0.1861 | -0.0185 | 0.0186  | 0.0070                     | 0.0003                  | 0.2950 | 0.2164                  | 0.2397                  | 1.24                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0002 | 0.0002  | 0.0002  | 0.9324                     | 0.0032                  | 0.0042 | 0.0022                  | 0.0026                  |                     |
| 50-60%                                    | 0.1192 | -0.0200 | 0.0156  | 0.0000                     | 0.0160                  | 0.2784 | 0.2118                  | 0.2630                  | 1.63                |
|                                           | $\pm$  | $\pm$   | $\pm$   | $\pm$                      | $\pm$                   | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                           | 0.0000 | 0.0000  | 0.0000  | 1.2132                     | 0.0002                  | 0.0000 | 0.0000                  | 0.0000                  |                     |

Table F.4: Fit parameters for data in Figure F.13, Figure F.14, Figure F.15, Figure F.16, Figure F.17, and Figure F.18.

|                                          | A      | $v_1^2$ | $v_2^2$ | $\frac{v_3^2}{v_5^2}$ | $\frac{v_4^2}{v_5^2}$ | $Y_J$  | $\sigma_{\Delta\eta,J}$ | $\sigma_{\Delta\phi,J}$ | $\chi^2/\text{NDF}$ |
|------------------------------------------|--------|---------|---------|-----------------------|-----------------------|--------|-------------------------|-------------------------|---------------------|
| <i>Au + Au</i> $\sqrt{s_{NN}} = 200$ GeV |        |         |         |                       |                       |        |                         |                         |                     |
| 0-12%                                    | 3.2879 | -0.0974 | -0.0138 | 0.2814                | 0.5273                | 0.1853 | 0.3629                  | 0.3183                  | 1.98                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0008 | 0.0012  | 0.0006  | 0.0025                | 0.0026                | 0.0012 | 0.0028                  | 0.0019                  |                     |
| 10-20%                                   | 2.2940 | -0.1093 | 0.0357  | 0.2962                | 0.5624                | 0.2173 | 0.3098                  | 0.2705                  | 1.10                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0022 | 0.0034  | 0.0014  | 0.0067                | 0.0061                | 0.0030 | 0.0047                  | 0.0036                  |                     |
| 20-30%                                   | 1.5698 | -0.0877 | 0.0589  | 0.2206                | 0.5698                | 0.2308 | 0.2962                  | 0.2735                  | 1.15                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0022 | 0.0033  | 0.0014  | 0.0066                | 0.0079                | 0.0029 | 0.0039                  | 0.0031                  |                     |
| 30-40%                                   | 1.0342 | -0.0557 | 0.0606  | 0.1279                | 0.5470                | 0.2476 | 0.2605                  | 0.2598                  | 1.10                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0017 | 0.0026  | 0.0012  | 0.0052                | 0.0113                | 0.0032 | 0.0031                  | 0.0031                  |                     |
| 40-50%                                   | 0.6488 | -0.0379 | 0.0446  | 0.0683                | 0.5528                | 0.2686 | 0.2567                  | 0.2537                  | 1.10                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0015 | 0.0023  | 0.0010  | 0.0047                | 0.0179                | 0.0034 | 0.0030                  | 0.0029                  |                     |
| 50-60%                                   | 0.3875 | -0.0207 | 0.0336  | 0.0138                | 0.4502                | 0.2861 | 0.2311                  | 0.2514                  | 1.09                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0005 | 0.0008  | 0.0005  | 0.0018                | 0.0308                | 0.0037 | 0.0024                  | 0.0028                  |                     |
| 60-70%                                   | 0.2126 | -0.0191 | 0.0230  | 0.0000                | 1.1770                | 0.2937 | 0.2263                  | 0.2466                  | 1.00                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0000 | 0.0000  | 0.0000  | 0.0000                | 0.8399                | 0.0001 | 0.0001                  | 0.0001                  |                     |
| 70-80%                                   | 0.1067 | -0.0209 | 0.0161  | 0.0000                | 1.1972                | 0.2983 | 0.2198                  | 0.2448                  | 1.12                |
|                                          | $\pm$  | $\pm$   | $\pm$   | $\pm$                 | $\pm$                 | $\pm$  | $\pm$                   | $\pm$                   |                     |
|                                          | 0.0002 | 0.0002  | 0.0002  | 0.1465                | 0.9002                | 0.0054 | 0.0027                  | 0.0031                  |                     |

Table F.5: Fit parameters for data in Figure F.13, Figure F.14, Figure F.15, Figure F.16, Figure F.17, and Figure F.18.



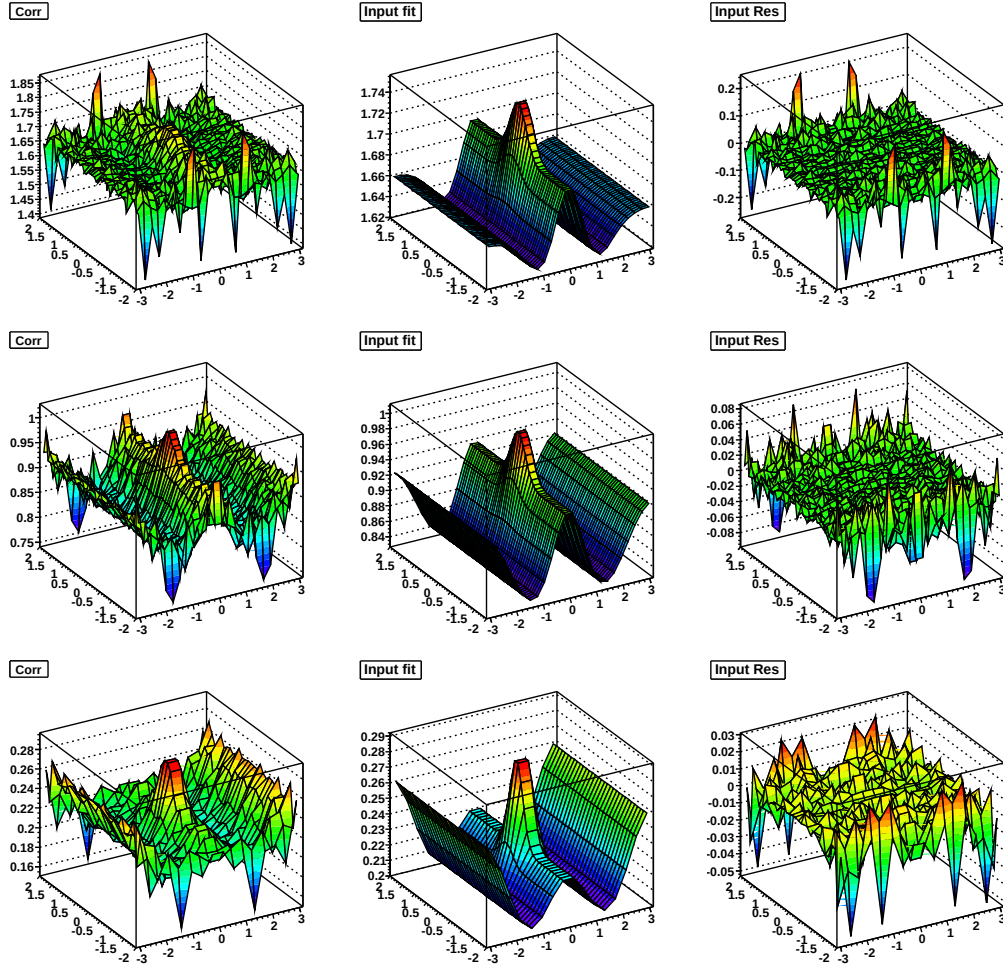


Figure F.14: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 0-10% (top), 10-40% (middle), and 40-80% (bottom)  $Au + Au \sqrt{s_{NN}} = 62 \text{ GeV}$



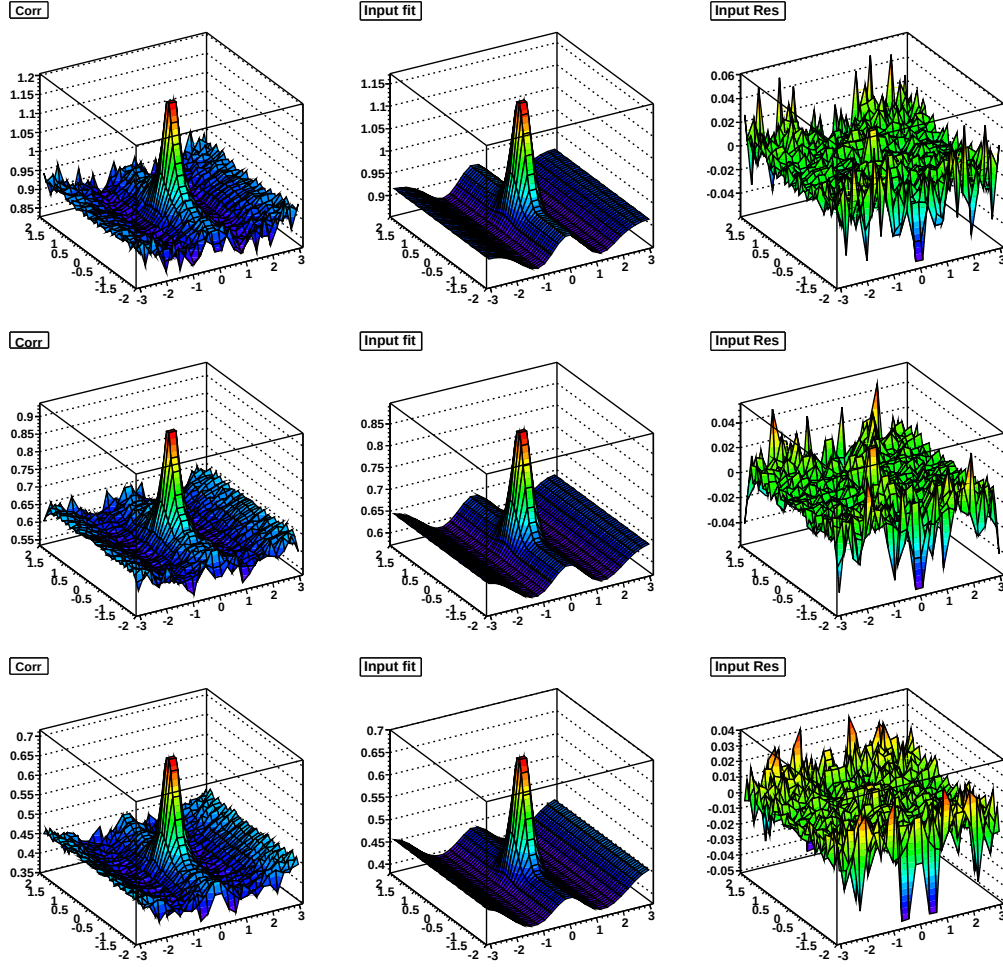


Figure F.15: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 0-10% (top), 10-20% (middle), and 20-30% (bottom)  $Cu + Cu$   $\sqrt{s_{NN}} = 200 \text{ GeV}$

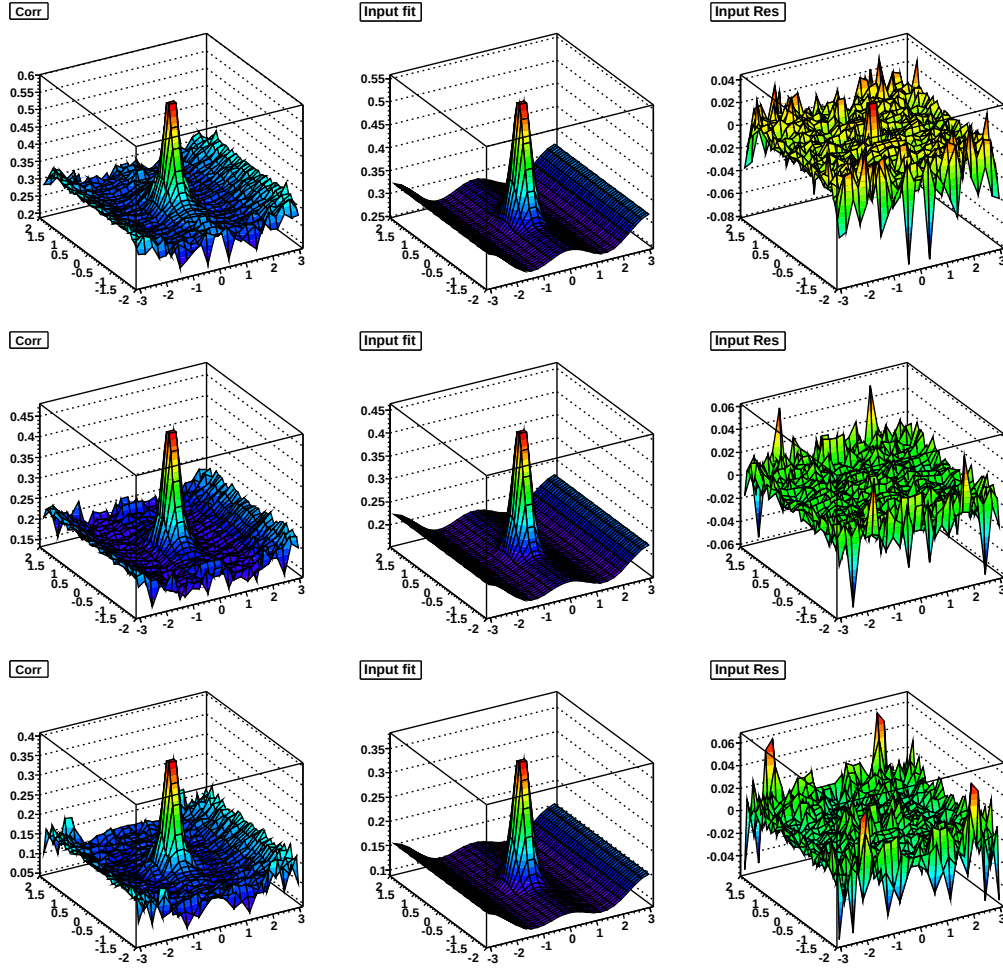


Figure F.16: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 30-40% (top), 40-50% (middle), and 50-60% (bottom)  $Cu + Cu$   $\sqrt{s_{NN}} = 200 \text{ GeV}$

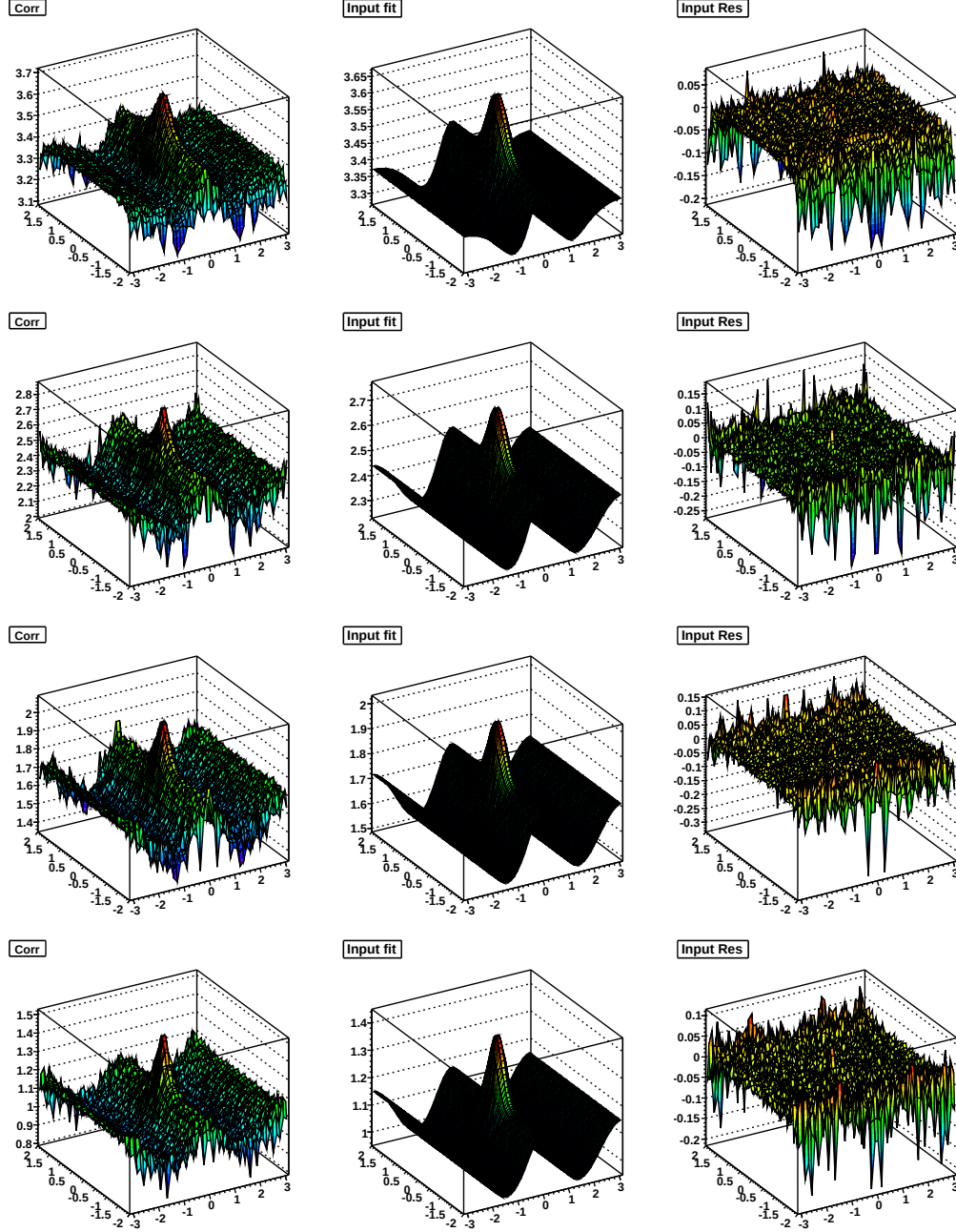


Figure F.17: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 0-12% (top), 10-20% (second down), 20-30% (third down), and 30-40% (bottom)  $Au + Au \sqrt{s_{NN}} = 200 \text{ GeV}$

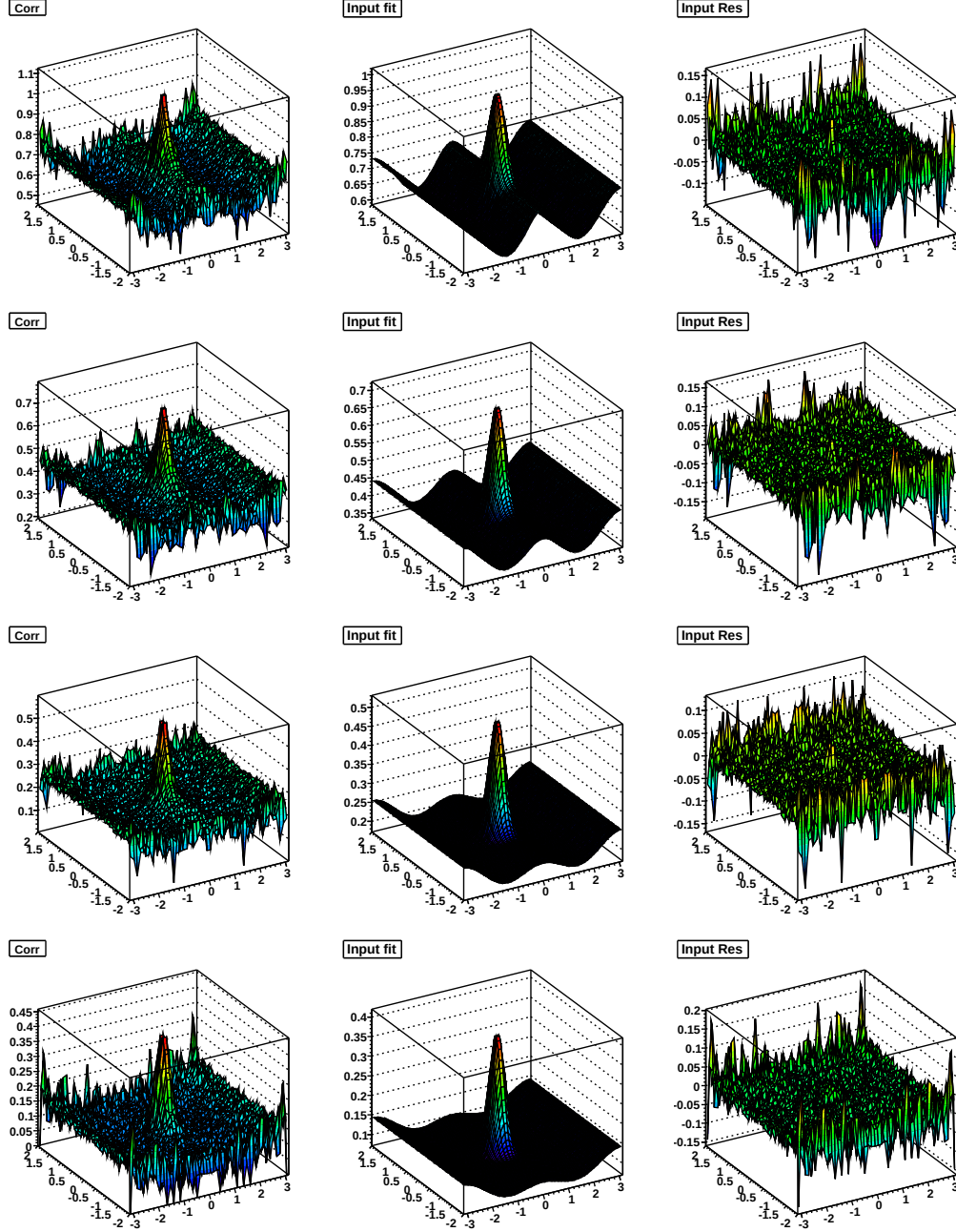


Figure F.18: 2D histogram (left), fit (middle), residuals (right) for  $1.5 \text{ GeV/c} < p_T^{\text{associated}} < p_T^{\text{trigger}}$  and  $3.0 < p_T^{\text{trigger}} < 6.0 \text{ GeV/c}$  in 40-50% (top), 50-60% (second down), 60-70% (third down), and 70-80% (bottom)  $Au + Au \sqrt{s_{NN}} = 200 \text{ GeV}$

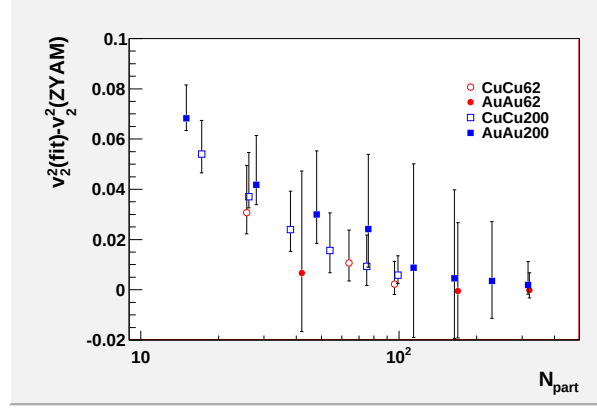


Figure F.19: The difference between the fit  $\tilde{v}_2^2$  and the standard  $v_2^{trigger} v_2^{assoc}$  used in the analysis.

| System         | $\sigma/\sigma_{\text{geo}}$<br>[%] | $\langle v_2^{\text{trigger}} \rangle$<br>[%] | $\langle v_2^{\text{assoc}} \rangle$<br>[%] | $\langle v_2^{\text{trigger}} \rangle \langle v_2^{\text{assoc}} \rangle$ | $v_2^{\text{fit}}$  |
|----------------|-------------------------------------|-----------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------|---------------------|
| <i>Cu + Cu</i> | 0-10                                | $8.5^{+1.4}_{-2.5}$                           | $7.3^{+1.9}_{-0.7}$                         | $0.0062^{0.0091}_{0.0040}$                                                | $0.0084 \pm 0.0005$ |
| 62.4 GeV       | 10-30                               | $11.6^{+0.5}_{-3.2}$                          | $10.4^{+0.3}_{-1.8}$                        | $0.0121^{0.0130}_{0.0072}$                                                | $0.0228 \pm 0.0005$ |
|                | 30-60                               | $14.5^{+1.0}_{-5.3}$                          | $11.8^{+0.3}_{-2.7}$                        | $0.0171^{0.0188}_{0.0084}$                                                | $0.0471 \pm 0.0010$ |
| <i>Au + Au</i> | 0-10                                | $8.5^{+0.3}_{-3.2}$                           | $7.6^{+0.3}_{-1.9}$                         | $0.0065^{0.0070}_{0.0030}$                                                | $0.0063 \pm 0.0003$ |
| 62.4 GeV       | 10-40                               | $17.5^{+0.3}_{-3.2}$                          | $15.0^{+0.3}_{-1.9}$                        | $0.0263^{0.0272}_{0.0187}$                                                | $0.0258 \pm 0.0002$ |
|                | 40-80                               | $21.2^{+0.3}_{-5.9}$                          | $18.6^{+0.3}_{-3.4}$                        | $0.0394^{0.0406}_{0.0233}$                                                | $0.0456 \pm 0.0008$ |
| <i>Cu + Cu</i> | 0-10                                | $8.0^{+0.7}_{-3.0}$                           | $8.8^{+0.1}_{-2.2}$                         | $0.0070^{0.0077}_{0.0033}$                                                | $0.0128 \pm 0.0002$ |
| 200 GeV        | 10-20                               | $10.5^{+0.6}_{-2.5}$                          | $11.3^{+0.0}_{-1.8}$                        | $0.0119^{0.0125}_{0.0076}$                                                | $0.0212 \pm 0.0002$ |
|                | 20-30                               | $11.2^{+0.7}_{-2.9}$                          | $12.5^{+0.1}_{-1.9}$                        | $0.0140^{0.0150}_{0.0088}$                                                | $0.0296 \pm 0.0003$ |
|                | 30-40                               | $10.7^{+0.9}_{-2.6}$                          | $13.0^{+0.1}_{-2.2}$                        | $0.0139^{0.0152}_{0.0087}$                                                | $0.0380 \pm 0.0004$ |
|                | 40-50                               | $11.2^{+2.2}_{-6.3}$                          | $12.6^{+0.5}_{-3.7}$                        | $0.0141^{0.0176}_{0.0044}$                                                | $0.0506 \pm 0.0006$ |
|                | 50-60                               | $10.3^{+0.6}_{-2.7}$                          | $12.3^{+0.0}_{-2.6}$                        | $0.0127^{0.0134}_{0.0074}$                                                | $0.0657 \pm 0.0010$ |
| <i>Au + Au</i> | 0-12                                | $8.5^{+2.1}_{-2.1}$                           | $7.3^{+1.5}_{-1.5}$                         | $0.0062^{0.0093}_{0.0037}$                                                | $0.0081 \pm 0.0001$ |
| 200 GeV        | 10-20                               | $14.6^{+2.0}_{-2.0}$                          | $13.0^{+1.2}_{-1.2}$                        | $0.0190^{0.0236}_{0.0149}$                                                | $0.0225 \pm 0.0001$ |
|                | 20-30                               | $17.8^{+2.1}_{-2.1}$                          | $16.5^{+1.2}_{-1.2}$                        | $0.0298^{0.0352}_{0.0240}$                                                | $0.0344 \pm 0.0001$ |
|                | 30-40                               | $18.9^{+2.3}_{-2.3}$                          | $18.1^{+1.4}_{-1.4}$                        | $0.0342^{0.0413}_{0.0277}$                                                | $0.0430 \pm 0.0001$ |
|                | 40-50                               | $11.5^{+2.3}_{-2.3}$                          | $19.0^{+2.5}_{-2.5}$                        | $0.0219^{0.0297}_{0.0152}$                                                | $0.0461 \pm 0.0002$ |
|                | 50-60                               | $10.1^{+2.3}_{-2.3}$                          | $17.6^{+2.8}_{-2.8}$                        | $0.0178^{0.0253}_{0.0115}$                                                | $0.0478 \pm 0.0003$ |
|                | 60-70                               | $8.5^{+2.2}_{-2.2}$                           | $15.4^{+2.9}_{-2.9}$                        | $0.0131^{0.0196}_{0.0079}$                                                | $0.0549 \pm 0.0006$ |
|                | 70-80                               | $6.6^{+1.9}_{-1.9}$                           | $12.8^{+2.8}_{-2.8}$                        | $0.0084^{0.0133}_{0.0047}$                                                | $0.0754 \pm 0.0012$ |

Table F.6:  $v_2$  values used in ZYAM method and from fit